

A blue triangle graphic pointing to the right, located to the left of the main title text.

FA report of JUNGGLIM GI56HAB03.2 Dark shadow MURA

Date 2026.02.12

Problem Statement



● General Information

Brand/Project	JUNGLIM	Defect Phenomenon	MURA														
AUO Model	GI56HAB03.2	Defect Station	LINE REJECT														
RMA No	HAND CARRY	Issue Date	2026/01/28														
AUO S/N	6R5B5EG4BTZZS0100	ES	莊淵翔														
Array/Module	<table><tr><th>ARRAY_FAB_ID</th><th>FEOL_FAB_ID</th><th>CF_FAB_ID</th><th>BEOL_FAB_ID</th><th>LCM_BEOL_FAB_ID</th><th>JI_FAB_ID</th><th>MODULE_FAB_ID</th></tr><tr><td>L5C</td><td>L5B</td><td>C5C</td><td>L5B</td><td>S01</td><td></td><td>S01</td></tr></table>			ARRAY_FAB_ID	FEOL_FAB_ID	CF_FAB_ID	BEOL_FAB_ID	LCM_BEOL_FAB_ID	JI_FAB_ID	MODULE_FAB_ID	L5C	L5B	C5C	L5B	S01		S01
ARRAY_FAB_ID	FEOL_FAB_ID	CF_FAB_ID	BEOL_FAB_ID	LCM_BEOL_FAB_ID	JI_FAB_ID	MODULE_FAB_ID											
L5C	L5B	C5C	L5B	S01		S01											

● Defect Panel S/N Picture

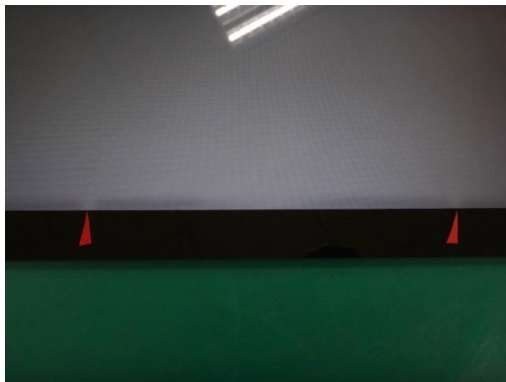


Problem Statement

- Describe the phenomenon at customer side

1) Kortek found dark shadow mura after aging 1h with 45.3 degrees.

- Defect Phenomenon



Summary of Root Cause Analysis



Item	Check item	Result	Summary
1	Mapping by Defect SN check (mother glass mapping)	No concentration on specific position	No concentration on specific position
2	Mapping by Defect SN check (process history)	Process is ok	Process is ok
3	HTO 60°C T1	Defect symptom can be found when aging 60°C & 1H but disappeared at the 25°C	Defect symptom can be duplicated.
4	Debonding Cover Glass and swap cell and backlight unit	Defect symptom is followed with Cell	Cell related
5	Check by Rotate POL (check cell gap issue)	No change	Cell gap is normal
6	Verify by adjust Vcom	Adjust Vcom, defect can not be found	No related Vcom
7	Check corresponding position from Cell condition and mura	Mura is focused on the smaller area of layout (line to line) spacing	Suspected it is related to sealant process.
8	Mapping curing process by defect SN	Focus on ASM line 20 (ASM 10 is ok.)	Focus on specific assmby line
9	Curing ratio check	Curing ratio in around 65%~85% (Target 90%)	Root cause is imcompletely curing ratio.

Note:

1. From SN mapping, no concentration on mother glass. Process history is all pass.
2. Defect symptom is followed with cell.
3. Electrical characteristic is OK.
4. After checking cell condition, we found the root cause is related to incomplete curing process due to insufficient curing ratio.

Root Cause Analysis

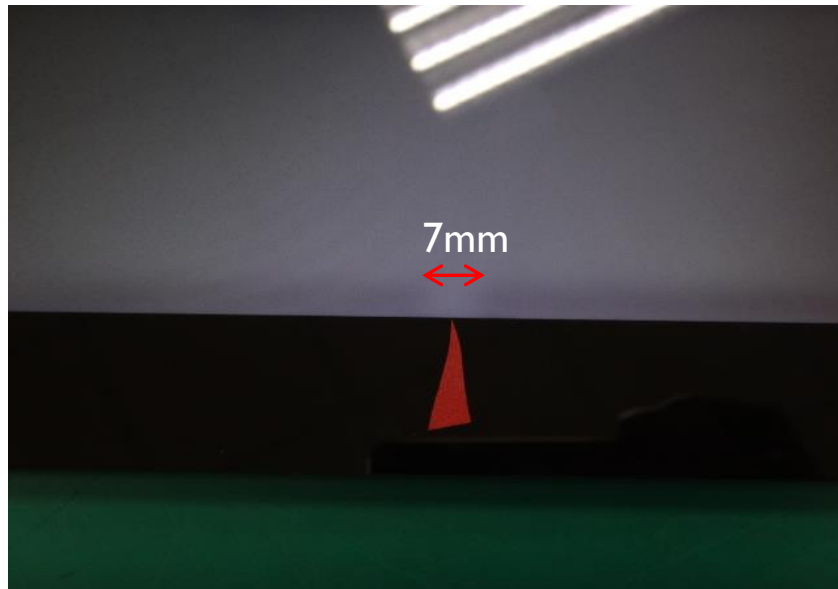


Replication experiment: (AUO LT FMA in Taiwan)

We use the defect panel from Junglim on Jan 28th to duplicate the dark shadow.

After increasing aging temperature from 50 to 60 degree , a black linear MURA appeared at the bottom—on the bottom side.

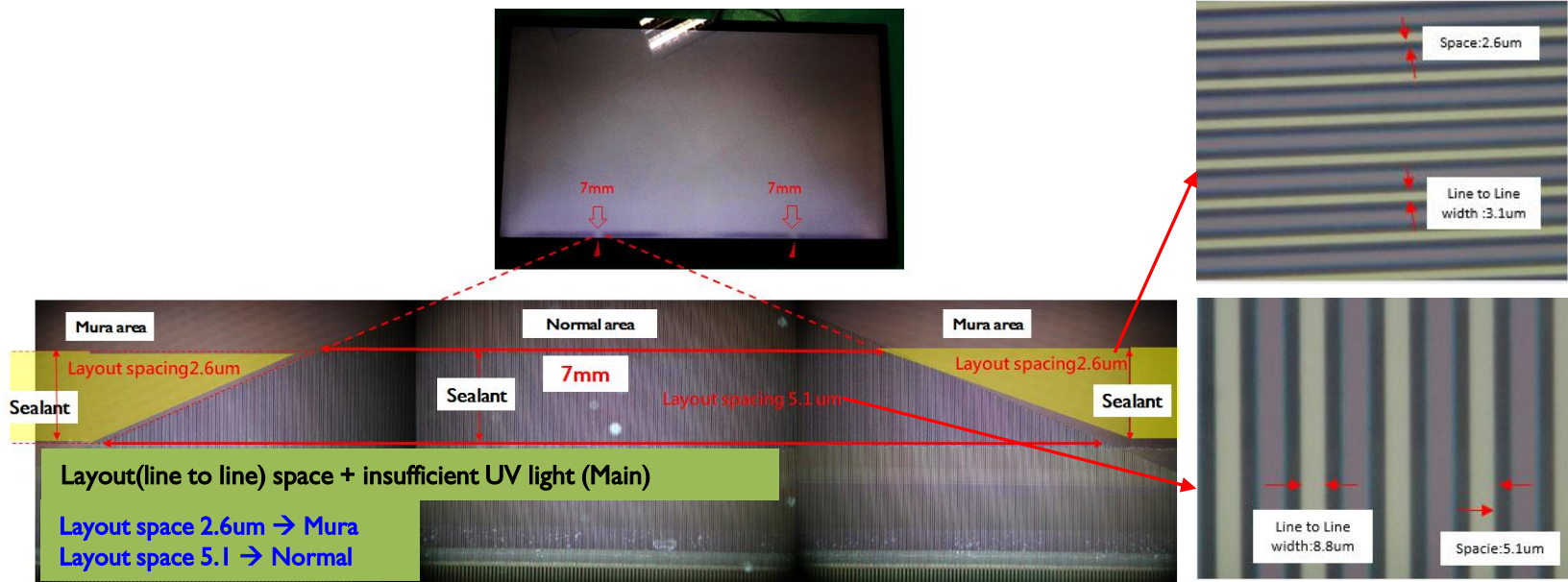
Inside the black linear MURA, there are two white notches located approximately 8.5 cm from the left and right edges of the AA area, and each notch is about 7 mm wide.



Root Cause Analysis

The corresponding position of Defect symptom and sealant area

1. From defect symptom, we found bottom mura is not continuous and there are two area (~7mm) which is normal.
2. From corresponding position, we found the layout spacing of mura area is 2.6um and the layout spacing of normal area is 5.1um. It means that when UV light exposure on the array layout area, the large spacing can be passing through more UV light. That's why we can find out mura is not continuous.
3. The layout spacing is fixed and meet design rule. The main impact is UV light energy is insufficient.



Root Cause Analysis

Curing ratio

1. After verifying the curing ratio in 6 area from defect panel JL provided, the curing ratio is insufficient.
2. The curing ratio of mura area (2&5) is under 80%. The curing ratio of area (1&3&4) is over 80%. The curing ratio of area 6 is over 90%. (Target : > 90%)
3. The main reason of mura is insufficient curing ratio.

周波数(cm-1)	820~860(Peak:830)	1402~1414(Peak:1408)	900~920(Peak:916)	Ac反應率(%)	Ep反應率(%)	Ac Total conversion (%)	Total conversion (%)
Name	Ref	Ac	Ep				
SWB66	0.697	0.826	0.057				
R-1-1(白區)	0.149	0.018	0.002	-0.750618156	1.038611451	-0.535910603	84.97% 84.97%
R-1-2(白區)	0.154	0.022	0.002	-1.070175439	1.027081967	-0.81850455	80.56% 80.56%
R-2-1(黑區)	0.172	0.026	0.001	-1.190534476	1.012172036	-0.926209694	77.96% 77.96%
R-2-2(黑區)	0.172	0.032	0.002	-1.696042432	1.017088285	-1.370466746	73.06% 73.06%
M-1-1(白區)	0.140	0.016	0.001	-0.656140351	1.022085561	-0.454753241	84.49% 84.49%
M-1-2(白區)	0.139	0.018	0.001	-0.876561908	1.016531893	-0.649390652	81.66% 81.66%
L-1-1(白區)	0.148	0.016	0.003	-0.566619251	1.076724686	-0.369417978	90.32% 90.32%
L-1-2(白區)	0.153	0.021	0.002	-0.988992088	1.029305043	-0.746796432	81.62% 81.62%
L-2-1(黑區)	0.176	0.030	0.002	-1.470095694	1.019714673	-1.17131845	75.68% 75.68%
L-2-2(黑區)	0.180	0.041	0.001	-2.300779727	1.006298399	-1.903930352	65.71% 65.71%
U-1-1(正常區)	0.103	0.012	0.005	-0.688298416	1.105268498	-0.473070386	91.59% 91.59%
U-1-2(正常區)	0.139	0.015	0.003	-0.56380159	1.077108126	-0.366892424	90.38% 90.38%



Root Cause Analysis



Assembly Process :

1. Used removed lamp on May 17th : From assembly process, AUO found fab use UV lamp which is removed lamp due to original lamp over 7000 hours.
2. The rule should not be allowed to use the removed lamp. But AUO fab didn't follow it.
3. Follow up: Modify the SOP. Feb. 26th.



Root Cause Analysis



Assembly Process Investigation

1. From the latest SN Junglim provided on Feb. 4th, the 44 pcs is from Assembly process line (ASM 20).
2. From the 448pcs SN of defect panels which provided by Junglim, we found 98% defect panels are from ASM 20.
3. The defect symptom of 4pcs Panel produced with ASM 10 can't be duplicated. ADP will exclude ASM 10 and define risk lot is ASM 20 on June 14th.
4. Currently AUO will stop using ASM20 and use ASM 10 to produce this product.

TFT_ASM_Day ▾	TFT_ASM_EQP ▾	NG	OK	總計
2022	#N/A		1	1
2025/4/21	CBASA010		30	30
2025/4/22	CBASA010		50	50
2025/4/23	CBASA010		1	1
2025/5/21	CBASA010	4	124	128
2025/6/14	CBASM020	187	51	238
總計		191	257	448

ASM 10 : No risk

ASM 20 : High risk

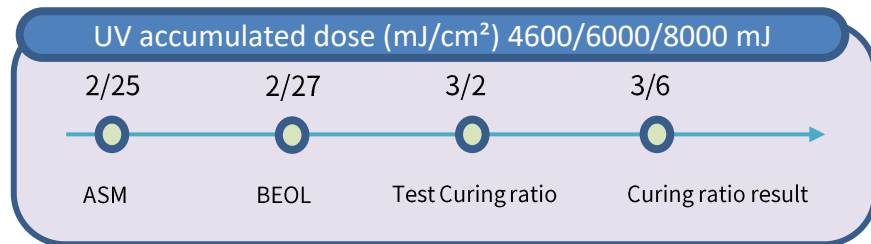
Assembly Process

1. Short term solution : Sort our stock and aging with 60 degree / 4 H to Junglim/Kortek

S01 HTO sorting by ASM line (use same week code with Kortek) : Focus on ASM 20 & 4 hours can be found.

ASM line	Q'ty	T2	T4	T8	T12	T24
#10	10	0%	0%	0%	0%	0%
#20	20	20%	100%	100%	100%	100%

2. Long term solution : Long term solution : Modify UV accumulated dose to make sure curing ratio is over 90%. ADP



Conclusion



● Conclusion and Countermeasure

1. From SN mapping, no concentration on mother glass. Process history is all pass.
2. Defect symptom is followed with cell and electrical characteristic is OK.
3. After checking cell condition, we found the root cause is related to incomplete curing process due to insufficient curing ratio.
4. After increasing aging temperature from 50 to 60 degree on defect panel from Junglim , a black linear MURA appeared at the bottom—on the bottom side.
5. From corresponding position, we found the layout spacing of mura area is 2.6um and the layout spacing of normal area is 5.1um. It means that when UV light exposure on the array layout area, the large spacing can be passing through more UV light. That's why we can find out mura is not continuous.
6. After verifying the curing ratio from defect panel JL provided, the curing ratio is insufficient.($<90\%$)
7. The insufficient UV curing is focused on ASM 20 and the risk lot is 354pcs. (Produced on June 14th)
8. Short term solution : Sort our stock and aging with 60 degree / 4 H to Junglim/Kortek
9. Long term solution : Modify UV accumulated dose to make sure curing ratio is over 90%.

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